

Physics SP015 Lecture Notes

Malaysian Matriculation Programme

Course Instructor

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1. Physical Quantities and Measurements

1.1. Introduction

Physics is the study of matter, energy, and their interactions. To describe physical phenomena precisely, we use *quantities*—measurable properties expressed with numbers and units. This chapter introduces the fundamental concepts needed to work with physical quantities systematically.

1.2. Dimensions and Dimensional Analysis

1.2.1. What is Dimension?

Definition

A **dimension** is a fundamental category of measurement that characterizes a physical quantity independently of its numerical value or units. The basic dimensions in mechanics are:

- **Length** [L]
- **Mass** [M]
- **Time** [T]
- **Temperature** [Θ]

All derived quantities can be expressed as combinations of these fundamental dimensions. For example:

$$\text{Velocity} : [LT^{-1}]$$

$$\text{Acceleration} : [LT^{-2}]$$

$$\text{Force} : [MLT^{-2}]$$

1.2.2. Dimensional Homogeneity

Definition

The principle of **dimensional homogeneity** states that all terms in a physical equation must have the same dimensions. An equation is dimensionally consistent only if both sides have identical dimensions.

Worked Example

Consider the kinematic equation: $v = u + at$

Check dimensional consistency:

- Left side: $[v] = [LT^{-1}]$ (velocity)
- Right side:

$$[u] = [LT^{-1}]$$

$$[at] = [LT^{-2}] \cdot [T] = [LT^{-1}]$$

- Conclusion: Both sides equal $[LT^{-1}]$ **Dimensionally consistent**

Important

A dimensionally consistent equation is not necessarily correct. While dimensional analysis can verify consistency, it cannot verify that numerical coefficients or functional forms are physically accurate.

1.3. Scalars and Vectors**1.3.1. Scalar Quantities****Definition**

A **scalar quantity** is completely described by a single numerical value (magnitude) and an appropriate unit. Scalars follow ordinary algebraic rules for addition and multiplication.

Examples of scalar quantities:

- Mass: 5 kg
- Temperature: 298 K
- Speed: $20 \frac{\text{m}}{\text{s}}$
- Energy: 100 J

1.3.2. Vector Quantities**Definition**

A **vector quantity** requires both magnitude and direction for complete description. Vectors obey special algebraic rules (vector addition) and are often represented by arrows or bold notation.

Examples of vector quantities:

- Displacement: 10 m at 30° to the horizontal
- Velocity: $20 \frac{\text{m}}{\text{s}}$ north
- Force: 50 N upward
- Acceleration: $9.81 \frac{\text{m}}{\text{s}^2}$ downward

1.3.3. Vector Resolution and Addition

In two dimensions, a vector **A** can be resolved into perpendicular components:

Key Formula

Horizontal component: $A_x = A \cos \theta$

Vertical component: $A_y = A \sin \theta$

where A is the magnitude and θ is the angle from the horizontal.

The resultant of two vectors **A** and **B** is found by:

$$R = \sqrt{(A_x + B_x)^2 + (A_y + B_y)^2}$$

Worked Example

Find the resultant of two forces: $F_1 = 30 \text{ N}$ at 0° (horizontal) and $F_2 = 40 \text{ N}$ at 90° (vertical).

Step 1: Resolve forces into components

$$F_{1x} = 30 \cos(0) = 30 \text{ N}$$

$$F_{1y} = 30 \sin(0) = 0$$

$$F_{2x} = 40 \cos(90) = 0$$

$$F_{2y} = 40 \sin(90) = 40 \text{ N}$$

Step 2: Sum components

$$R_x = F_{1x} + F_{2x} = 30 + 0 = 30 \text{ N}$$

$$R_y = F_{1y} + F_{2y} = 0 + 40 = 40 \text{ N}$$

Step 3: Calculate resultant magnitude

$$\begin{aligned} R &= \sqrt{R_x^2 + R_y^2} \\ &= \sqrt{30^2 + 40^2} \\ &= \sqrt{900 + 1600} \\ &= \sqrt{2500} = 50 \text{ N} \end{aligned}$$

$$\text{Direction: } \tan \theta = \frac{R_y}{R_x} = \frac{40}{30} \Rightarrow \theta = 53.1^\circ$$

1.4. Significant Figures and Uncertainty

1.4.1. Significant Figures

Definition

Significant figures are the digits in a measurement that carry meaningful information about the precision of the measurement. Rules for counting significant figures:

1. Non-zero digits are always significant
2. Zeros between non-zero digits are significant
3. Leading zeros are not significant
4. Trailing zeros are significant only after a decimal point

1.4.2. Rules for Calculations

Key Formula

Addition and Subtraction: Round result to the least number of decimal places in any number

Example: $12.50 \text{ m} + 1.25 \text{ m} = 13.75 \text{ m}$ (limited to 2 decimal places)

Multiplication and Division: Round result to the least number of significant figures in any number

Example: $12.5 \times 1.25 = 15.6$ (limited to 3 sig. figs.)

Worked Example

Calculate the area of a rectangle with length 15.25 cm and width 8.1 cm.

$$\begin{aligned} A &= l \times w \\ &= 15.25 \text{ cm} \times 8.1 \text{ cm} \\ &= 123.525 \text{ cm}^2 \end{aligned}$$

The width (8.1 cm) has only 2 significant figures, so the answer must be rounded to 2 sig. figs.:

$$A = 1.2 \times 10^2 \text{ cm}^2 = 120 \text{ cm}^2$$

1.4.3. Uncertainty Analysis

Definition

Uncertainty (or error) in a measurement represents the range of values within which the true value is likely to lie. Systematic uncertainties arise from measurement apparatus limitations; random uncertainties from unpredictable fluctuations.

For combined measurements, uncertainties add in quadrature:

$$\delta Q = \sqrt{\left(\frac{\partial Q}{\partial x} \delta x\right)^2 + \left(\frac{\partial Q}{\partial y} \delta y\right)^2 + \dots}$$

Worked Example

A velocity is calculated as $v = \frac{s}{t}$ where $s = 10.0(2)$ m and $t = 2.0(1)$ s. Find the velocity with uncertainty.

Nominal value:

$$v = \frac{10.0}{2.0} = 5.0 \frac{\text{m}}{\text{s}}$$

Partial derivatives:

$$\begin{aligned}\frac{\partial v}{\partial s} &= \frac{1}{t} = 0.5 \\ \frac{\partial v}{\partial t} &= -\frac{s}{t^2} = -2.5\end{aligned}$$

Uncertainty:

$$\begin{aligned}\delta v &= \sqrt{(0.5 \times 0.2)^2 + (2.5 \times 0.1)^2} \\ &= \sqrt{(0.1)^2 + (0.25)^2} \\ &= \sqrt{0.01 + 0.0625} \\ &= 0.26 \frac{\text{m}}{\text{s}}\end{aligned}$$

Result: $v = 5.0(3) \frac{\text{m}}{\text{s}}$ (rounded to 1 sig. fig. in uncertainty)

1.5. Summary of Key Points

- Dimensions characterize physical quantities; dimensional analysis checks consistency
- Scalars require only magnitude; vectors require magnitude and direction
- Significant figures reflect measurement precision
- Uncertainties propagate through calculations and should be reported with results

1.6. Exercises

[title=Exercise 1] The density of an object is defined as $\rho = \frac{m}{V}$, where m is mass and V is volume. Determine the dimension of density.

[title=Exercise 2] Two forces are applied to an object: $\mathbf{F}_1 = 12 \text{ N}$ due east and $\mathbf{F}_2 = 16 \text{ N}$ due north. Find the magnitude and direction of the resultant force.

[title=Exercise 3] A measurement yields 2.340 m. How many significant figures does this value contain? If this is added to 1.2 m, what is the result with correct sig. figs.?

[title=Exercise 4] The power dissipated in a resistor is given by $P = I^2 R$, where I is current (in amperes) and R is resistance (in ohms). Show that the dimensions of power are $[ML^2T^{-3}]$.

2. Kinematics of Linear Motion

2.1. Linear Motion Basics

Definition

Kinematics is the branch of mechanics that describes motion without reference to the forces causing it. **Linear motion** is motion along a straight line.

Key concepts in linear kinematics:

- **Position** (s or x): location of the object
- **Displacement** (Δs): change in position (vector)
- **Distance**: total path length traveled (scalar)
- **Velocity**: rate of change of position (vector)
- **Speed**: magnitude of velocity (scalar)
- **Acceleration**: rate of change of velocity (vector)

2.1.1. Velocity and Speed

Velocity and Speed Definitions

Average velocity: $\bar{v} = \frac{\Delta s}{\Delta t}$

Instantaneous velocity: $v = \frac{ds}{dt}$

Average speed: $\bar{c} = \frac{\text{total distance}}{\text{total time}}$

Note. Velocity is a vector (direction matters); speed is the scalar magnitude of velocity. For motion in one dimension, velocity can be positive or negative to indicate direction.

2.1.2. Acceleration

Acceleration

Average acceleration: $\bar{a} = \frac{\Delta v}{\Delta t}$

Instantaneous acceleration: $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$

Important

Zero acceleration does not mean zero velocity. An object moving at constant velocity (e.g., a car cruising at $30 \frac{\text{m}}{\text{s}}$) has zero acceleration despite non-zero velocity.

2.2. Uniformly Accelerated Motion

When acceleration is constant, we can use the *kinematic equations*:

Kinematic Equations (Constant Acceleration)

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{u + v}{2} \cdot t$$

where:

- u = initial velocity
- v = final velocity
- a = acceleration
- t = time interval
- s = displacement

Worked Example

A car accelerates uniformly from rest with acceleration $2 \frac{\text{m}}{\text{s}^2}$ for 5 s. Find:

[label=()]

1. The final velocity
2. The displacement

Given: $u = 0, a = 2 \frac{\text{m}}{\text{s}^2}, t = 5 \text{ s}$

Part (a): Final velocity

Using $v = u + at$:

$$\begin{aligned} v &= 0 + 2 \times 5 \\ &= 10 \frac{\text{m}}{\text{s}} \end{aligned}$$

Part (b): Displacement

Using $s = ut + \frac{1}{2}at^2$:

$$\begin{aligned} s &= 0 \times 5 + \frac{1}{2} \times 2 \times 5^2 \\ &= 0 + 1 \times 25 \\ &= 25 \text{ m} \end{aligned}$$

[title=Exercise 5] A ball is thrown vertically upward with initial velocity $20 \frac{\text{m}}{\text{s}}$. Taking $g = 10 \frac{\text{m}}{\text{s}^2}$ downward, find:

[label=()]

1. Time to reach maximum height
2. Maximum height
3. Time to return to starting point

References

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A. Mathematical Formulas and Constants

A.1. SI Base Units

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Table A.1.. SI Base Units

A.2. Common Physics Constants

Constant	Symbol	Value
Gravitational acceleration	g	$9.81 \frac{\text{m}}{\text{s}^2}$
Speed of light	c	$3.00 \times 10^8 \frac{\text{m}}{\text{s}}$
Planck constant	h	$6.626 \times 10^{-34} \text{J} \cdot \text{s}$
Boltzmann constant	k_B	$1.381 \times 10^{-23} \frac{\text{J}}{\text{K}}$
Avogadro constant	N_A	$6.022 \times 10^{23} \frac{1}{\text{mol}}$

Table A.2.. Common Physics Constants